

SETTING UP A KICK START SERVER

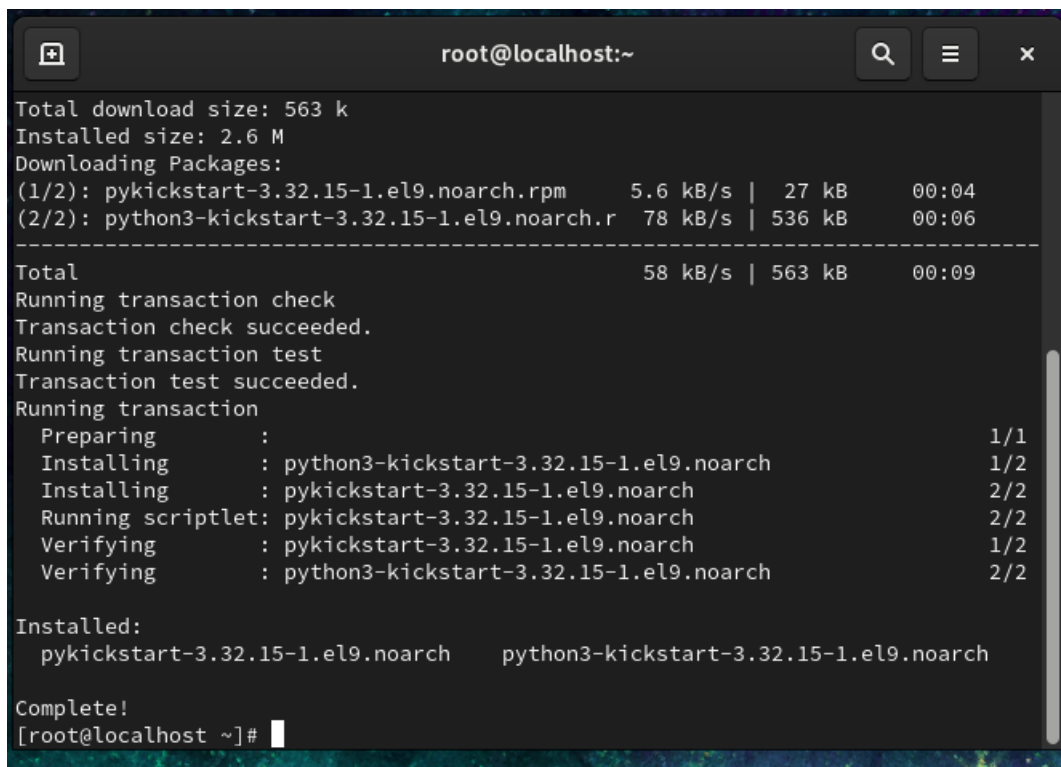
TASKS

1.Install the kickstart package on your Linux server and verify the installation by running the appropriate command.

Install Kickstart Package

Step 1: Install Package

`dnf install pykickstart -y`

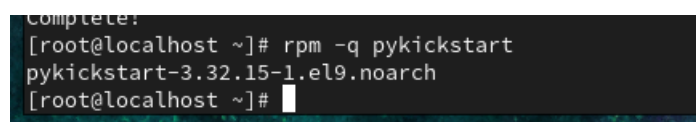


```
root@localhost:~  
Total download size: 563 k  
Installed size: 2.6 M  
Downloading Packages:  
(1/2): pykickstart-3.32.15-1.el9.noarch.rpm 5.6 kB/s | 27 kB 00:04  
(2/2): python3-kickstart-3.32.15-1.el9.noarch.r 78 kB/s | 536 kB 00:06  
-----  
Total 58 kB/s | 563 kB 00:09  
Running transaction check  
Transaction check succeeded.  
Running transaction test  
Transaction test succeeded.  
Running transaction  
  Preparing      : 1/1  
  Installing    : python3-kickstart-3.32.15-1.el9.noarch 1/2  
  Installing    : pykickstart-3.32.15-1.el9.noarch 2/2  
  Running scriptlet: pykickstart-3.32.15-1.el9.noarch 2/2  
  Verifying     : pykickstart-3.32.15-1.el9.noarch 1/2  
  Verifying     : python3-kickstart-3.32.15-1.el9.noarch 2/2  
  
Installed:  
  pykickstart-3.32.15-1.el9.noarch  python3-kickstart-3.32.15-1.el9.noarch  
  
Complete!  
[root@localhost ~]#
```

Step
2:
Verify

Installation

`rpm -q pykickstart`



```
Complete:  
[root@localhost ~]# rpm -q pykickstart  
pykickstart-3.32.15-1.el9.noarch  
[root@localhost ~]#
```

2.Create a basic kickstart configuration file (ks.cfg) that automates the installation of CentOS or Rocky Linux with your preferred timezone, root password, and a user account named 'student'.

Create Kickstart Configuration File

Step 1: Create File

```
mkdir -p /var/www/html/kickstart
```

```
vi /var/www/html/kickstart/ks.cfg
```

Press:

i

Paste the following:

```
#version=RHEL9
```

```
text
```

```
lang en_US.UTF-8
```

```
keyboard us
```

```
timezone Asia/Kolkata
```

```
rootpw Root@123
```

```
user --name=student --password=Student@123
```

```
network --bootproto=dhcp
```

```
firewall --enabled
```

```
selinux --enforcing
```

```
bootloader --location=mbr
```

```
clearpart --all --initlabel
```

```
autopart
```

```
reboot
```

Save:

Esc

:wq

```
[root@localhost ~]# ls /var/www/html/kickstart
ks.cfg
```

Step 2: Validate Kickstart File

ksvalidator /var/www/html/kickstart/ks.cfg

```
[root@localhost ~]# vi /var/www/html/kickstart/ks.cfg
[root@localhost ~]# ksvalidator /var/www/html/kickstart/ks.cfg
[root@localhost ~]#
```

3.Set

up an HTTP server (using Apache or nginx) to serve your kickstart file, and confirm you can access ks.cfg from another machine on the network via a browser or curl.

Serve Kickstart File Through Apache

Step 1: Apache Running

systemctl status httpd

If close:

systemctl start httpd
systemctl enable httpd

Step 2: Check File

ls -l /var/www/html/kickstart

```
Jul 22 21:20:51 localhost.localdomain httpd[14758]: S
[root@localhost ~]# ls -l /var/www/html/kickstart
total 4
-rw-r--r--. 1 root root 277 Jul 22 21:47 ks.cfg
[root@localhost ~]#
```

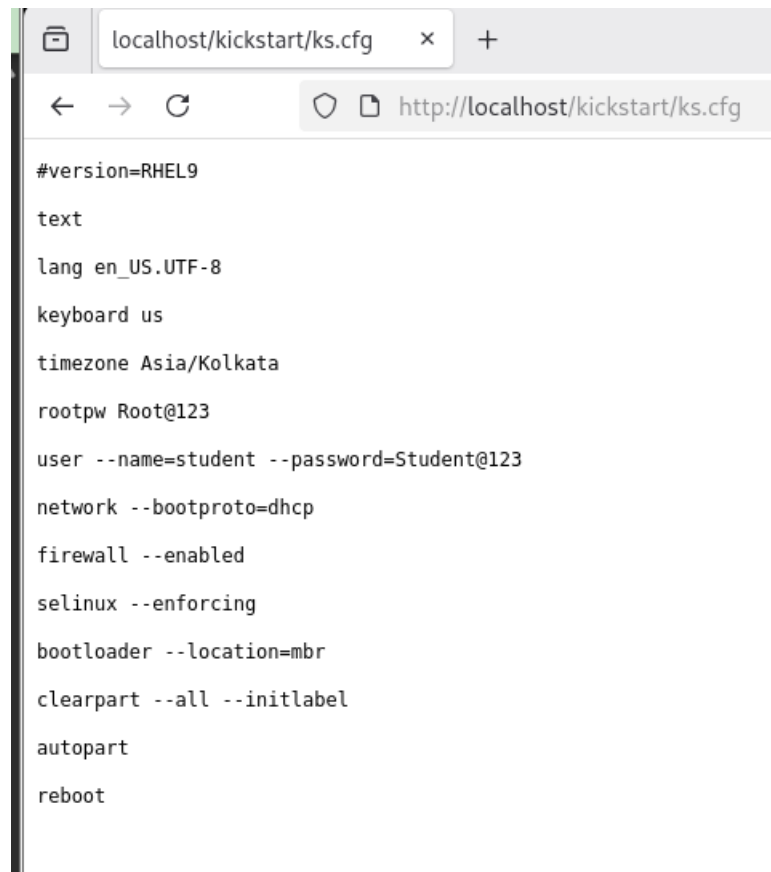
Step 3: Browser Test

Linux Browser:

http://localhost/kickstart/ks.cfg

or

http://192.168.231.128/kickstart/ks.cfg



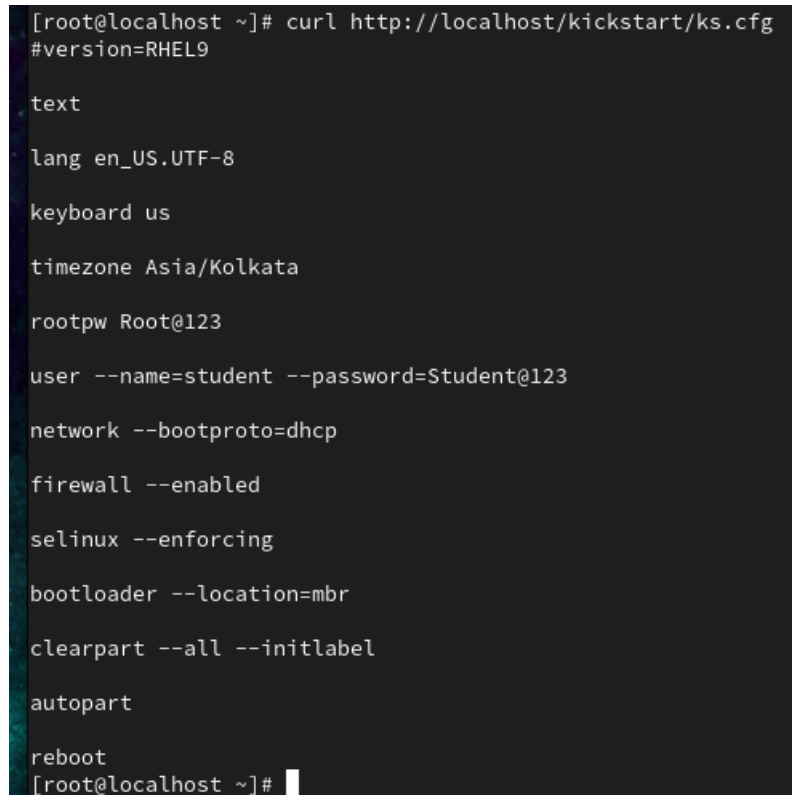
A screenshot of a web browser window. The address bar shows 'http://localhost/kickstart/ks.cfg'. The page content is a text file with the following configuration for a RHEL9 kickstart:

```
#version=RHEL9
text
lang en_US.UTF-8
keyboard us
timezone Asia/Kolkata
rootpw Root@123
user --name=student --password=Student@123
network --bootproto=dhcp
firewall --enabled
selinux --enforcing
bootloader --location=mbr
clearpart --all --initlabel
autopart
reboot
```

Step 4: Curl Test

curl

<http://localhost/kickstart/ks.cfg>



A screenshot of a terminal window. The prompt is '[root@localhost ~]#'. The command 'curl http://localhost/kickstart/ks.cfg' has been executed, and the output is the same kickstart configuration as seen in the browser screenshot:

```
[root@localhost ~]# curl http://localhost/kickstart/ks.cfg
#version=RHEL9
text
lang en_US.UTF-8
keyboard us
timezone Asia/Kolkata
rootpw Root@123
user --name=student --password=Student@123
network --bootproto=dhcp
firewall --enabled
selinux --enforcing
bootloader --location=mbr
clearpart --all --initlabel
autopart
reboot
[root@localhost ~]#
```

4. Modify your kickstart file to automatically install the GNOME desktop environment and add a post-install script that creates a text file

**/home/student/welcome.txt with the message 'Welcome to your automated Linux setup!'.

Hint: Use the %post section in your ks.cfg for post-install commands.**

Modify Kickstart File

Open file

```
vi /var/www/html/kickstart/ks.cfg
```

Add this in bottom:

```
%packages
```

```
@^Server with GUI
```

```
%end
```

```
%post
```

```
echo "Welcome to your automated Linux setup!" > /home/student/welcome.txt
```

```
%end
```

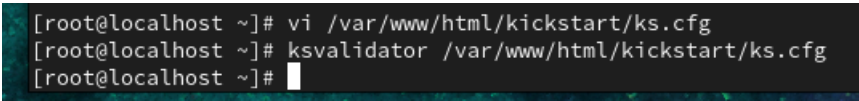
Save:

Esc

:wq

Validate Again

```
ksvalidator /var/www/html/kickstart/ks.cfg
```



```
[root@localhost ~]# vi /var/www/html/kickstart/ks.cfg
[root@localhost ~]# ksvalidator /var/www/html/kickstart/ks.cfg
[root@localhost ~]#
```

View File

```
cat /var/www/html/kickstart/ks.cfg
```

```
[root@localhost ~]# cat /var/www/html/kickstart/ks.cfg
#version=RHEL9

text

lang en_US.UTF-8

keyboard us

timezone Asia/Kolkata

rootpw Root@123

user --name=student --password=Student@123

network --bootproto=dhcp

firewall --enabled

selinux --enforcing

bootloader --location=mbr

clearpart --all --initlabel

autopart

reboot

%packages
@^Server with GUI
%end

%post
echo "Welcome to your automated Linux setup!" > /home/student/welcome.txt
%end
[root@localhost ~]#
```